

Deletion of node in Binary Search Tree (BST)

For a given BST, there may arise following three situations for deletion a node. The cases are:

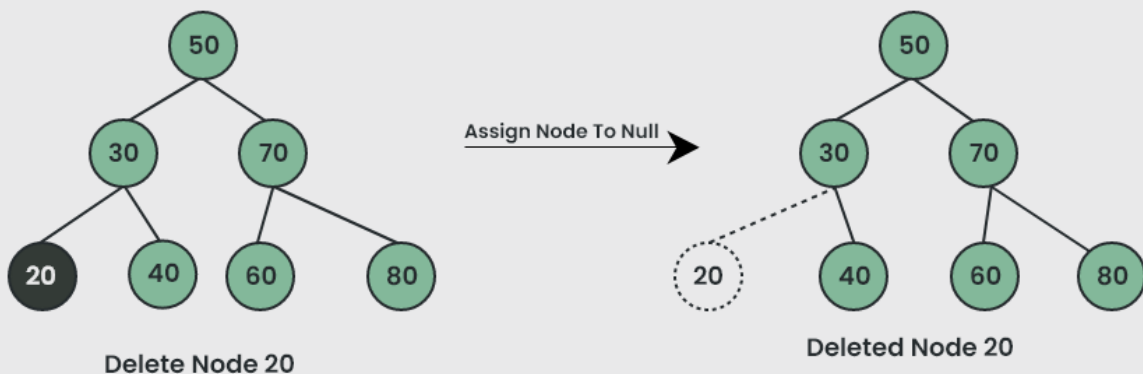
- 1) delete a leaf node
- 2) delete a node having single child
- 3) delete a node having two child

Case-I: Delete a leaf node.

In this case, simply remove the leaf node from the BST. The remaining tree is the required BST.

For example,

Case 1 : Delete A Leaf Node In BST



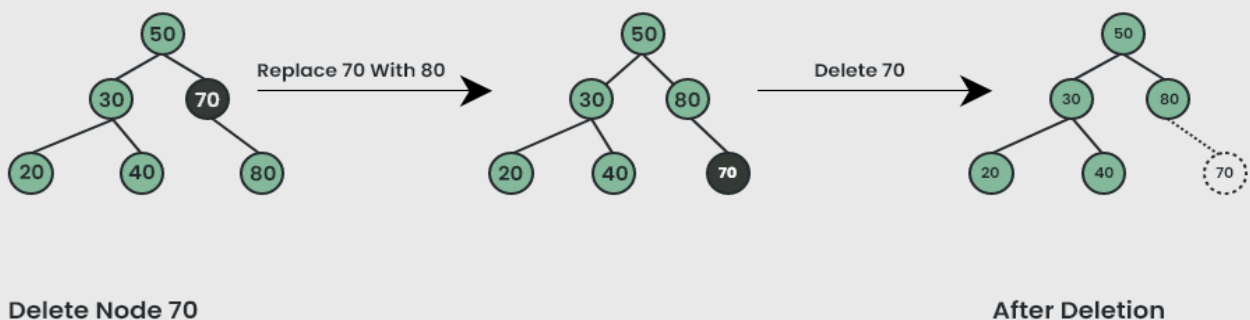
Deletion In BST



Case-II: Delete a node having single child.

In this case, the deleted node is to be replaced with its child. For example,

Case 2: Delete A Node With Single Child In BST



Deletion In BST



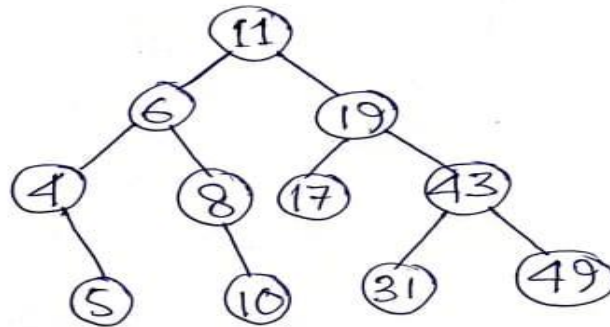
Case-III: Delete a node having two child.

There are two methods to delete such a node.

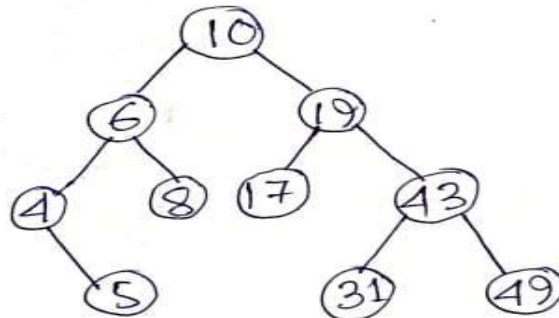
Method-I (Inorder predecessor)- Inorder predecessor is the largest element of the left subtree. In this case, the node is to be replaced with the largest element of the left subtree. For example, see the last figure.

Method-II (Inorder successor)- Inorder successor is the smallest element of the right subtree. In this case, the node is to be replaced with the smallest element of the right subtree. For example,

Delete 11 from the BST below-



Method-I (Inorder predecessor) -



Method-II (Inorder successor) -

